

● COAL — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. First Coal Mine — Raniganj, Damodar River (1774)
2. Coal Mining Timeline (Raniganj→NCDC→Nationalisation→CIL)
3. Coalfield Map Trap — Karanpura is in Jharkhand (Not Elsewhere)
4. Gondwana System — Main Source of Coal (99%+)
5. Gondwana Provides 90%+ of India's Coal
6. Indian Coal Characteristics — High Ash, Low Sulphur
7. Gondwana Rocks — Most Important Due to 90%+ Coal Reserves
8. India Self-Sufficient in Coking Coal for 3 Decades
9. Coal Reserves — 293.50 Billion Tonnes Assertion (Disputed)
10. Jharkhand & Odisha — 50%+ of Total Coal Reserves
11. China #1 in Coal Production; India Has Vast Reserves (JH, CG)
12. Historical Coal Ranking — Undivided Bihar > Undivided MP (1999)
13. Two Leading Coal States — Jharkhand & Chhattisgarh (Historical)
14. Coal Production Descending Order (Jharkhand>Odisha>MP>Chhattisgarh, Historical)
15. Odisha — Largest Coal Producing State (24% Share)
16. Coal Reserves Descending Order (Odisha>Jharkhand>Chhattisgarh>WB>MP)
17. Major Coal Reserve States — Jharkhand, Chhattisgarh, Odisha (All Correct)
18. Odisha — Largest Coal Reserves & Largest Producer
19. Coal Reserves Ranking (Historical: Jharkhand>Odisha>Chhattisgarh>WB>MP)
20. Coal Reserves Ranking (2022: Jharkhand>Odisha>Chhattisgarh>WB>MP)
21. Coal Production 2014 Ranking (Chhattisgarh>Jharkhand>Odisha>Maharashtra)
22. Chota Nagpur Industrial Region — Grew from Coal Discovery
23. Jharkhand's Coal Production Share — 16.72% (Disputed)
24. Three Leading Coal Producers (Historical Chhattisgarh>Odisha>Jharkhand)
25. Two Largest Coal Producers 2008-09 — Chhattisgarh & Odisha
26. Korba Coalfield — Chhattisgarh (Hasdeo River Basin)
27. Coal Valley-Field Match (Damodar-Barakar, Son-Umaria, Godavari-Singareni, Mahanadi-Talcher)
28. Coalfield-Valley Match (Talcher-Mahanadi, Karanpura-Damodar, Singroli-Son, Singareni-Godavari)
29. Talcher — Odisha's Famous Coal Region (Angul District)
30. Chota Nagpur's Coal Production Contribution — ~64% (2021-22)
31. Coal Mines in Jharkhand — Jharia (Not Jamshedpur/Ranchi/Lohardaga)
32. Maximum Coal Reserve Region — Raniganj (WB Portion)
33. Jharia — Coal Deposits, Dhanbad District
34. Jharia Coalfield — Located in Jharkhand
35. Lignite — Inferior Coal Variety (40-55% Carbon, TN Leader)
36. Largest Lignite Deposit — Tamil Nadu (Neyveli)
37. Neyveli — India's Largest Lignite Coalfield
38. Tamil Nadu — Largest Lignite Reserve State
39. Bengal-Jharkhand Coalfields — Major Source for NW States Rail Traffic
40. Birsampur — Coal Mining, Surajpur District, Chhattisgarh
41. Coalfield-State Match (Karanpura-Jharkhand, Singareni-Telangana, Neyveli-TN, Korba-Chhattisgarh)
42. Coal Import Reasons — Power Plants + Coking Coal Needs (Not Policy)
43. Coalbed Methane & Shale Gas — Both Sources Exist in India
44. Indian Coal Industry Problems — Quality, Washeries, Imports, Prices
45. Maximum Coal Use — Energy Production

- 46. Lignite NOT a Major Reserve in Jharkhand (TN, Rajasthan, Gujarat Are)
- 47. Damodar River Valley — Most Important for Gondwana-Age Coal
- 48. Coalfield-State Match (Rajmahal-Jharkhand, Sohagpur-MP, Bismarck-CG, Raniganj-WB)
- 49. Northeast Coalfield Match (Makum-Assam, Namchik-Arunachal, Giridih-Jharkhand, Sohagpur-MP)
- 50. Namchik-Namphuk Coalfields — Arunachal Pradesh (Changlang District)
- 51. Coalfield-State Match (Jhilmili-Chhattisgarh, Karanpura-Jharkhand, Rampur Himgir-Odisha, Sohagpur-MP)
- 52. NE Coalfield Match (Ranji Valley-Sikkim, Tandur-Telangana, Warora-Maharashtra, Lalmatia-Jharkhand)
- 53. Coal Resource Match (Hutar-Jharkhand, Ib River-Odisha, Tatapani-Ramkola-Chhattisgarh, Umaria-MP)
- 54. Underground Coal Combustion — Jharkhand (Jharia Coalfield)
- 55. Coal Controller's Organization (CCO) — Statistics, Monitoring, Objections

1. First Coal Mine — Raniganj, Damodar River (1774)

- India ki pehli coal mine Raniganj mein Damodar river ke tat par khodi gayi thi (1774 mein). [Q1]

2. Coal Mining Timeline (Raniganj→NCDC→Nationalisation→CIL)

- India ka coal mining ~220 saal purana hai — chronological order: **Raniganj mein pehla coal production (1774)**, NCDC establish (1956), coal mines ka nationalisation (1971-72 coking, 1973 non-coking), Coal India Ltd (CIL) establish. [Q2]

3. Coalfield Map Trap — Karanpura is in Jharkhand (Not Elsewhere)

- Satpura, Son Valley, Korba coalfields sahi jagah dikhaye gaye — '**Karanpura**' galat dikhaya gaya tha (**Karanpura asal mein Jharkhand mein hai**). [Q3]

4. Gondwana System — Main Source of Coal (99%+)

- India ke coal deposits mainly **Gondwana sediments (Peninsular India ke eastern-central parts)** mein concentrated hain — Assam-Sikkim mein bhi milte hain — 99% se zyada coal Gondwana se aata hai. [Q4, Q5, Q6]

5. Gondwana Provides 90%+ of India's Coal

- Gondwana rock system **90% se zyada India ka coal deta hai**. [Q5, Q6]

6. Indian Coal Characteristics — High Ash, Low Sulphur

- Indian coal 2 areas mein milta hai: Gondwana region (high-grade, kam ash, high thermal power) aur Kalp/Tertiary region (low-quality, sulphur-rich) — **Gondwana coal ki characteristics: high ash content, low sulphur content** — 'low ash fusion temperature' Indian coal ki characteristic **NAHI** hai. [Q7]

7. Gondwana Rocks — Most Important Due to 90%+ Coal Reserves

- Gondwana rocks ko India ki sabse important rock system maana jaata hai kyunki **90% se zyada India ke coal reserves inme paaye jaate hain**. [Q8]

8. India Self-Sufficient in Coking Coal for 3 Decades

- Energy Policy report (Planning Commission) ke mutabik: **petroleum reserves 22 saal mein khatam ho jaayenge, gas 30 saal mein, coal 80 saal mein** — India coking coal mein agle 3 dashak tak self-sufficient

rahega — Indian Minerals Yearbook 2022 ke mutabik 1200m depth tak 361411 million tonnes coal resources hain. [Q9]

9. Coal Reserves — 293.50 Billion Tonnes Assertion (Disputed)

- Indian Minerals Yearbook 2022 ke mutabik coal reserves 361411 million tonnes hain (Assertion mein diya 293.50 billion tonnes figure purana/galat hai) — **Jharkhand aur Odisha mein desh ke kul coal reserves ka aadhe se zyada hissa hai (Reason sahi hai)** — Public Service Commission ne answer (b) maana tha, par sahi answer (d) hona chahiye (Assertion galat, Reason sahi) — disputed question hai. [Q10]

10. Jharkhand & Odisha — 50%+ of Total Coal Reserves

- 1 April 2022 tak **Odisha (88105 MT), Jharkhand (86660 MT), Chhattisgarh (74192 MT) sabse zyada geological coal resources wale states hain — Odisha+Jharkhand milke desh ke kul coal deposits ka 48.4% hain** — total geological coal resources 361411 million tonnes. [Q11]

11. China #1 in Coal Production; India Has Vast Reserves (JH, CG)

- 2021 mein **China world coal-lignite production ka ~50% share rakhta tha, India dusre number par (9.0%), phir USA (7.88%), Indonesia (7.80%), Australia (6.02%), Russia (5.48%)** — India ke paas Jharkhand-Chhattisgarh jaise states mein vishaal coal reserves hain (sab statements sahi hain). [Q12]

12. Historical Coal Ranking — Undivided Bihar > Undivided MP (1999)

- 1999 mein **undivided Bihar (Jharkhand included) ka coal production undivided Madhya Pradesh (Chhattisgarh included) se zyada tha — sahi order: Bihar, MP, WB** — abhi (2021-22, P) top-3: Odisha, Chhattisgarh, MP. [Q13]

13. Two Leading Coal States — Jharkhand & Chhattisgarh (Historical)

- Question ke waqt **Jharkhand aur Chhattisgarh do leading coal-producing states the — Indian Minerals Yearbook 2022 (2021-22, P) ke mutabik ab Odisha aur Chhattisgarh top-2 hain.** [Q14]

14. Coal Production Descending Order (Jharkhand>Odisha>MP>Chhattisgarh, Historical)

- 2007-08 ke data ke mutabik top coal producers: **Jharkhand, Chhattisgarh, Odisha, Madhya Pradesh — Indian Minerals Yearbook 2022 (2021-22, P) ke top-5: Odisha > Chhattisgarh > Madhya Pradesh > Jharkhand > Telangana** — coal ko 'Black Diamond' aur 'Fuel Mineral' kaha jaata hai. [Q15]

15. Odisha — Largest Coal Producing State (24% Share)

- Indian Minerals Yearbook 2022 ke mutabik **Odisha ~24% share ke saath coal ka sabse bada producer state hai, Chhattisgarh dusre number par hai.** [Q16, Q17]

16. Coal Reserves Descending Order (Odisha>Jharkhand>Chhattisgarh>WB>MP)

- Coal (proved) reserves ke descending order (1 April 2022 tak): **Odisha > Jharkhand > Chhattisgarh > West Bengal > Madhya Pradesh** — diye gaye options mein se koi bhi match nahi karta, isliye disputed question hai. [Q18]

17. Major Coal Reserve States — Jharkhand, Chhattisgarh, Odisha (All Correct)

- Jharkhand, Chhattisgarh, aur Odisha sab teeno India ke major coal-reserve states hain (in mein se koi bhi ek nahi, teeno sahi hain). [Q19]

18. Odisha — Largest Coal Reserves & Largest Producer

- Indian Minerals Yearbook 2022 ke mutabik Odisha total coal reserves aur production (2021-22, P) dono mein sabse aage hai — 88105 million tonnes reserves. [Q20]

19. Coal Reserves Ranking (Historical: Jharkhand>Odisha>Chhattisgarh>WB>MP)

- 1 April 2021 tak coal (proved) reserves ka chronological order: Jharkhand > Odisha > Chhattisgarh > West Bengal > Madhya Pradesh. [Q21]

20. Coal Reserves Ranking (2022: Jharkhand>Odisha>Chhattisgarh>WB>MP)

- Indian Minerals Yearbook 2022 (1 April 2022) ke mutabik: Jharkhand (53245 MT) > Odisha (48573 MT) > Chhattisgarh (32053 MT) > West Bengal (17234 MT) > Madhya Pradesh (14052 MT). [Q22]

21. Coal Production 2014 Ranking (Chhattisgarh>Jharkhand>Odisha>Maharashtra)

- Coal Statistics 2014-15 ke mutabik: Chhattisgarh (22.5%) > Jharkhand (20.0%) > Odisha (19.9%) > Maharashtra — Indian Minerals Yearbook 2022 (2021-22, P) ke top-5: Odisha > Chhattisgarh > MP > Jharkhand > Telangana. [Q23]

22. Chota Nagpur Industrial Region — Grew from Coal Discovery

- Chota Nagpur industrial region (Jharkhand, North Odisha, WB mein extended) Damodar Valley mein coal ki khoj ki wajah se develop hua — heavy metallurgical industry ke liye famous hai. [Q24]

23. Jharkhand's Coal Production Share — 16.72% (Disputed)

- Indian Minerals Yearbook 2022 ke mutabik 2021-22 (P) mein total coal production mein Jharkhand ka share 16.72% tha — diye gaye options (40/45/47/49%) mein se koi bhi sahi nahi tha, disputed question hai. [Q25]

24. Three Leading Coal Producers (Historical Chhattisgarh>Odisha>Jharkhand)

- 2008-09 mein Chhattisgarh (20.7%), Odisha (20.0%), Jharkhand (19.5%) coal production mein top-3 the — abhi (2021-22, P): Odisha (24%), Chhattisgarh (20%), Madhya Pradesh (18%). [Q26]

25. Two Largest Coal Producers 2008-09 — Chhattisgarh & Odisha

- 2008-09 mein Chhattisgarh aur Odisha coal ke do sabse bade producers the. [Q27]

26. Korba Coalfield — Chhattisgarh (Hasdeo River Basin)

- Korba coalfield Korba district, Chhattisgarh mein Hasdeo river ke basin mein hai. [Q28]

27. Coal Valley-Field Match (Damodar-Barakar, Son-Umaria, Godavari-Singareni, Mahanadi-Talcher)

- Match: Damodar Valley — Barakar, Son Valley — Umaria, Godavari Valley — Singareni, Mahanadi Valley — Talcher. [Q29]

28. Coalfield-Valley Match (Talcher-Mahanadi, Karnapura-Damodar, Singroli-Son, Singareni-Godavari)

- Match: Talcher — Mahanadi Valley, Karnapura — Damodar Valley, Singroli — Son Valley, Singareni — Godavari Valley. [Q30]

29. Talcher — Odisha's Famous Coal Region (Angul District)

- Talcher Angul district, Odisha ka prominent coal region hai — Mahanadi Coalfield Ltd (Coal India Limited ki subsidiary) manage karti hai. [Q31, Q32]

30. Chota Nagpur's Coal Production Contribution — ~64% (2021-22)

- Chota Nagpur plateau (Jharkhand, Chhattisgarh, Odisha, WB mein extended) ka India ke total coal production mein 64.04% [2021-22 (P)] contribution hai. [Q33]

31. Coal Mines in Jharkhand — Jharia (Not Jamshedpur/Ranchi/Lohardaga)

- Jharia coalfield Dhanbad district, Jharkhand mein hai — zyadatar coking coal reserves yahan hain — Jamshedpur iron/steel ke liye, Lohardaga bauxite ke liye, Ranchi heavy machines ke liye famous hain. [Q34]

32. Maximum Coal Reserve Region — Raniganj (WB Portion)

- Indian Minerals Yearbook 2022 (1 April 2022) ke mutabik Raniganj (West Bengal portion, 26357 MT total reserves) mein sabse zyada coal reserves hain — Jharia (19533 MT), Singrauli-MP (16919 MT), Korba (12976 MT) se zyada. [Q35]

33. Jharia — Coal Deposits, Dhanbad District

- Jharia (Jharkhand) mein mukhya roop se coal paaya jaata hai. [Q36]

34. Jharia Coalfield — Located in Jharkhand

- Jharia coalfield Jharkhand mein sthith hai. [Q37]

35. Lignite — Inferior Coal Variety (40-55% Carbon, TN Leader)

- Lignite 40-55% carbon wali inferior coal variety hai (Assertion mein diya 35-40% figure GALAT hai) — Jharkhand lignite ka largest producer NAHI hai (Tamil Nadu hai, Reason bhi galat hai) — mainly Tamil Nadu, Gujarat, Rajasthan mein produce hoti hai. [Q38]



EXAM TRAP — Lignite India ka largest producer Tamil Nadu hai, Jharkhand NAHI

36. Largest Lignite Deposit — Tamil Nadu (Neyveli)

- 2021-22 (P) mein lignite production 47 million tonnes tha (~25% badha pichhle saal se) — **Tamil Nadu ~50% production, Gujarat ~28%, Rajasthan ~22%** — **Tamil Nadu mein India ka sabse bada lignite deposit hai.** [Q39]

37. Neyveli — India's Largest Lignite Coalfield

- Neyveli (Tamil Nadu) mein **India ka sabse bada lignite coal reserve hai** — **reserves/resources aur production dono mein Tamil Nadu top par hai.** [Q40]

38. Tamil Nadu — Largest Lignite Reserve State

- Minerals Yearbook 2022 ke mutabik India mein total 46204.16 million tonnes lignite reserves/resources hain — **Tamil Nadu mein sabse zyada lignite reserves hain.** [Q41]

39. Bengal-Jharkhand Coalfields — Major Source for NW States Rail Traffic

- Coal ki inter-state movement railways ke through hone wale inter-state traffic ka major component hai (Assertion sahi) — **kyunki Bengal-Jharkhand coalfields north-western states ko coal supply ka major source hain (Reason bhi sahi hai, sahi explanation hai).** [Q42]

40. Bistrampur — Coal Mining, Surajpur District, Chhattisgarh

- Bistrampur (Surajpur district, Chhattisgarh) **coal mining ke liye famous hai.** [Q43]

41. Coalfield-State Match (Karanpura-Jharkhand, Singareni-Telangana, Neyveli-TN, Korba-Chhattisgarh)

- Match: **Karanpura — Jharkhand, Singareni — Andhra Pradesh (ab Telangana), Neyveli — Tamil Nadu, Korba — Chhattisgarh.** [Q44]

42. Coal Import Reasons — Power Plants + Coking Coal Needs (Not Policy)

- India coal import karta hai kyunki **India ke zyadatar power plants coal-based hain aur unhe desh ke andar se paryapt coal supply nahi milti, aur steel companies ko coking coal ki zaroorat hoti hai jo import karna padta hai** — 'apne coal reserves bachaane ki policy' galat reason hai. [Q45]

43. Coalbed Methane & Shale Gas — Both Sources Exist in India

- Coalbed methane (coal seam se nikalti hai, 90%+ pure methane) aur shale gas (fine-grained sedimentary rocks se, 86%+ methane) dono unconventional energy sources hain — **India mein dono ke abundant sources hain (Indo-Gangetic plain, Assam, Gujarat, Rajasthan, coastal areas mein shale gas bhi milti hai)** — 'shale gas sirf propane-butane hai' aur 'India mein shale gas nahi milti' dono statements **GALAT** hain. [Q46]

44. Indian Coal Industry Problems — Quality, Washeries, Imports, Prices

- Indian coal industry ki sab 4 problems sahi hain: **poor quality/movement bottlenecks, washeries ki low utilization capacity, coking coal import par badhti nirbharta, administered prices.** [Q47]

45. Maximum Coal Use — Energy Production

- Coal ka sabse zyada use energy production mein hota hai — India mein ek indispensable energy source hai. [Q48]

46. Lignite NOT a Major Reserve in Jharkhand (TN, Rajasthan, Gujarat Are)

- Lignite (brown coal) high moisture, low ash, low heating value wala hota hai — **Jharkhand mein lignite ka major reserve NAHI hai** — Tamil Nadu, Rajasthan, Gujarat mein hai. [Q49]

⚠ EXAM TRAP — Lignite ka major reserve — Tamil Nadu/Rajasthan/Gujarat mein hai, Jharkhand mein NAHI

47. Damodar River Valley — Most Important for Gondwana-Age Coal

- Gondwana-age coal reserves ke hisaab se **Damodar river valley sabse important hai** — Jharia/Dhanbad/Bokaro (Jharkhand), Raniganj coalfield (WB, Jharkhand tak bhi extended) is valley mein aate hain. [Q50]

48. Coalfield-State Match (Rajmahal-Jharkhand, Sohagpur-MP, Bistrampur-CG, Raniganj-WB)

- Match: **Rajmahal — Jharkhand+Bihar, Sohagpur — MP+Chhattisgarh, Bistrampur — Chhattisgarh, Raniganj — WB+Jharkhand.** [Q51]

49. Northeast Coalfield Match (Makum-Assam, Namchik-Arunachal, Giridih-Jharkhand, Sohagpur-MP)

- Match: **Makum — Assam, Namchik — Arunachal Pradesh, Giridih — Jharkhand (best-quality coal), Sohagpur — MP — Namchik-Namphuk Arunachal Pradesh ka ekmatra coalfield hai.** [Q52]

50. Namchik-Namphuk Coalfields — Arunachal Pradesh (Changlang District)

- Namchik-Namphuk coal fields **Changlang district, Arunachal Pradesh** mein hain. [Q53]

51. Coalfield-State Match (Jhilmili-Chhattisgarh, Karanpura-Jharkhand, Rampur Himgir-Odisha, Sohagpur-MP)

- Match: **Jhilmili — Chhattisgarh, Karanpura — Jharkhand, Rampur Himgir — Odisha, Sohagpur — Madhya Pradesh+Chhattisgarh.** [Q54]

52. NE Coalfield Match (Ranjit Valley-Sikkim, Tandur-Telangana, Warora-Maharashtra, Lalmatia-Jharkhand)

- Match: **Ranjit Valley — Sikkim, Tandur — Telangana, Warora — Maharashtra, Lalmatia — Jharkhand.** [Q55]

53. Coal Resource Match (Hutar-Jharkhand, Ib River-Odisha, Tatapani-Ramkola-Chhattisgarh, Umaria-MP)

- Indian Minerals Yearbook 2022 ke mutabik match: **Hutar — Jharkhand, Ib River — Odisha, Tatapani-Ramkola — Chhattisgarh, Umaria — Madhya Pradesh.** [Q56]

54. Underground Coal Combustion — Jharkhand (Jharia Coalfield)

- Underground coal ka combustion (jalna) **Jharkhand state mein ho raha hai — Jharia coalfield mein.** [Q57]

55. Coal Controller's Organization (CCO) — Statistics, Monitoring, Objections

- CCO (1916 mein establish, pehle 'Coal Commissioner' kehlata tha) ka role: **Coal Statistics ka major source hai, captive coal/lignite blocks ki development monitor karta hai, coal-bearing areas ke acquisition-notification par objections sunta hai — 'coal companies ko end-users ko time par coal deliver karvana' CCO ki responsibility NAHI hai (Coal Bearing Areas Act 1957 ke tahat).** [Q58]